AI-Assignment 6

Name - > Anubhav Sharma

Roll No -> IIT2024234

Question 1->

1. CSP Formulation

A Constraint Satisfaction Problem (CSP) is represented as CSP = (X, D, C).

Variables

X = {A, B, C, D, E}, where each variable represents one map region.

Domains

D(A) = D(B) = D(C) = D(D) = D(E) = {Red, Green, Blue}.

Constraints

Neighboring regions must have different colors:

- $A \neq B$
- $A \neq C$
- $B \neq C$
- $B \neq D$
- $B \neq E$
- $C \neq E$
- $D \neq E$

2. Ordinary Backtracking Search

Variable order: A → B → C → D → E. Values are tried in the order Red → Green → Blue.

Step	Variable	Value Tried	Result
1	A	Red	Valid
2	B	Red	Rejected — conflicts with A
3	B	Green	Valid
4	C	Red	Rejected — conflicts with A
5	C	Green	Rejected — conflicts with B
6	C	Blue	Valid
7	D	Red	Valid
8	E	Red	Rejected — conflicts with D
9	E	Green	Rejected — conflicts with B
10	E	Blue	Rejected — conflicts with C and D
11	D	Green	Rejected — conflicts with B
12	D	Blue	Valid
13	E	Red	Valid

3. Backtracking Point

After A=Red, B=Green, C=Blue, D=Red, E has no valid color because E is adjacent to B, C, and D, which use Green, Blue, and Red respectively. Therefore, backtracking is required.

Backtrack from D=Red. D=Green is invalid because D is adjacent to B=Green. D=Blue is valid, after which E=Red is valid.

4. One Valid Final Assignment

A = Red, B = Green, C = Blue, D = Blue, E = Red.

Neighbor Pair	Colors	Valid?
A-B	Red / Green	Yes
A-C	Red / Blue	Yes
B-C	Green / Blue	Yes
B-D	Green / Blue	Yes
B-E	Green / Red	Yes
C-E	Blue / Red	Yes
D-E	Blue / Red	Yes

5. MRV Heuristic

MRV (Minimum Remaining Values) selects the variable with the fewest legal values remaining. Initially all variables have 3 values, so there is a tie. Using the degree of the variable as a tie-breaker, B is selected because it has the most neighbors.

Step	Variable	Assignment	Effect
1	B	Red	A,C,D,E → {Green, Blue}
2	C	Green	A → {Blue}; E → {Blue}; D → {Green, Blue}
3	A	Blue	A fixed
4	E	Blue	E fixed
5	D	Green	D fixed

MRV finds the valid solution B=Red, C=Green, A=Blue, E=Blue, D=Green without backtracking.

6. Comparison

Method	Backtracking	Observation
Ordinary Backtracking	Yes (1 dead-end)	Reaches a dead end at E and then changes D.
MRV	No	More constrained variables are handled earlier, avoiding the dead end.

Conclusion: MRV can substantially reduce unnecessary search. For this tie-breaking order, ordinary backtracking requires one backtracking event, whereas MRV finds a solution directly.

Question 2->

1. CSP Formulation

Variables

X = {AI, DBMS, OS, CN}. Each course must be assigned exactly one time slot.

Domains

Course	Initial Domain
AI	{1, 2, 3}
DBMS	{1, 2, 4}
OS	{2, 3, 4}
CN	{1, 3, 4}

Constraints

- AI ≠ DBMS
- AI ≠ OS
- DBMS ≠ CN
- OS ≠ CN

Additional unary constraints are already represented by the domains.

2. Ordinary Backtracking Search

Variable order: AI → DBMS → OS → CN. Values are tried in ascending order.

Step	Course	Value	Result
1	AI	1	Valid
2	DBMS	1	Rejected — conflicts with AI
3	DBMS	2	Valid
4	OS	2	Valid
5	CN	1	Valid

Final solution: AI=1, DBMS=2, OS=2, CN=1. No backtracking is required; DBMS=1 is simply rejected and the next value is tried.

3. Complete Assignment Sequence

AI=1 → DBMS=1 (reject) → DBMS=2 → OS=2 → CN=1.

4. MRV Selection

Initially all four courses have three possible values. Therefore there is a tie. Using the original course order as the tie-breaker, AI is selected first.

5. MRV + Forward Checking

Forward Checking removes values from the domains of unassigned variables that would violate a constraint with the newly assigned variable.

Step	MRV Variable	Value Assigned	Updated Domains After Forward Checking	Backtracking

1	AI	1	DBMS={2,4}; OS={2,3,4}; CN={1,3,4}	No
2	DBMS	2	OS={2,3,4}; CN={1,3,4}	No
3	OS	2	CN={1,3,4}	No
4	CN	1	All variables assigned	No

6. Detailed Forward Checking

Step 1 — Select AI using MRV. Assign AI=1. Since AI cannot share a slot with DBMS or OS, remove 1 from their domains: DBMS={2,4}, OS={2,3,4}. CN is unchanged.

Step 2 — DBMS now has the fewest remaining values (2), so MRV selects DBMS. Assign DBMS=2. CN cannot use 2, but 2 was not in CN's domain, so CN remains {1,3,4}.

Step 3 — OS and CN both have three values. Use the original order as the tie-breaker and select OS. Assign OS=2. Since CN has no 2 in its domain, its domain remains {1,3,4}.

Step 4 — CN is the only unassigned course. Assign CN=1. All constraints are satisfied.

7. Final Valid Timetable

Time Slot	**Course(s)**
1	AI, CN
2	DBMS, OS
3	—
4	—

8. Constraint Verification

Constraint	**Check**	**Status**
AI ≠ DBMS	1 ≠ 2	✓
AI ≠ OS	1 ≠ 2	✓
DBMS ≠ CN	2 ≠ 1	✓
OS ≠ CN	2 ≠ 1	✓
AI ∈ {1,2,3}	1 is allowed	✓
DBMS ∈ {1,2,4}	2 is allowed	✓
OS ∈ {2,3,4}	2 is allowed	✓
CN ∈ {1,3,4}	1 is allowed	✓